Assignment 1 725/0800 Chen ~~Bai~~ Baizheng

Question 1: Knapsack with two resource constraints

(a) Let x_i indicate selection of item i, the integer program:

$$\max \sum_{i=1}^{n} v_i x_i$$
$$\text{subject to } \sum_{i=1}^{n} a_i x_i \le A_i$$
$$\sum_{i=1}^{n} b_i x_i \le B_i$$
$$x_i \in \{0,1\} \ (i=1,\dots,n)$$

Relax $x_i \in \{0,1\}$ to $0 \le x_i \le 1$, each integer solution is LP-feasible,

~~hence~~ $\therefore OPT \le OPT_{LP}$

(b) Either selection single item is optimal: $OPT = 1$

$$(1+\delta)(x_1+x_2) \le 2$$
$$\Rightarrow x_1^* = x_2^* = \frac{1}{1+\delta}, \quad OPT_{LP} = \frac{2}{1+\delta}$$

Rounding down gives $(0,0)$ of value 0.

(c) If an extreme point x has 3 fractional coordinates

$$\Rightarrow \Sigma_i a_i d_i = \Sigma_i b_i d_i = 0$$

For small $\varepsilon > 0$, $x \pm \varepsilon d$ are distince feasible points with midpoint x

$\therefore$ At most 2 coordinates are fractional

(d) Let $F = \{i : 0 < x_i^* < 1\}$, then $|F| \le 2$ and $v_i \le v_j$

$$\Rightarrow OPT \le OPT_{LP} = v(S) + \sum_{i \in F} v_i^* x_i^*$$
$$\le v(S) + 2v_j \le 3\max\{v(S), v_j\}$$

The reture value is at least $\frac{OPT}{3}$

Question 2: Weighted vertex cover in a hypergraph

(a) Let x_i indicate selection of v, the integer program:

$$\min \sum_{v\in V} c_v x_v$$

$$\text{s.t. } \sum_{v\in e} x_v \geq 1 \quad (e\in E)$$

$$x_v \in \{0,1\} \quad (v\in V)$$

$x_v\in\{0,1\}$ to $0\leq x_v\leq 1$, every integer solution is LP-~~fea~~ feasible

$\Rightarrow OPT_{LP}\leq OPT$

(b) If a hyperedge e misses C, every $x_v^*<\frac{1}{r}$.

$\Rightarrow \sum_{v\in e} x_v^* < \frac{|e|}{r} \leq 1$

contradicting the LP constraint $\sum_{v\in e} x_v^* \geq 1$.

$\Rightarrow$ C is a cover

(c) $v\in C$, $1\leq r x_v^*$

$\Rightarrow c(C)=\sum_{v\in C} c_v \leq r\sum_{v\in C} c_v x_v^* \leq r OPT_{LP} \leq r OPT$

solving the LP and rounding take polynomial time. An r-approximation

(d) A cover omits at most $r-1$ ~~vertex~~ vertices, or an r-edge is uncovered

$\Rightarrow OPT = n-(r-1)$

Let $x_v=\frac{1}{r}$, $LP=\frac{n}{r}$, summing all edge constraints with each vertex appearing $\binom{n-1}{r-1}$ times

$\Rightarrow \binom{n-1}{r-1}\sum_{v\in V} x_v \geq \binom{n}{r}$, $\sum_{v\in V} x_v \geq \frac{n}{r}$

$\Rightarrow OPT_{LP}=\frac{n}{r}$, $\frac{OPT}{OPT_{LP}}=\frac{r[n-(r-1)]}{n} \Rightarrow r$ $(n\to\infty, r \text{ fixed})$

The gap concerns this LP, not the hardness of all algorithms.

Question 5 & 6 : Online matching with capacities

(a) Greedy: $v_1 \to a$, $v_2 \to a$; reject v_3, value = 2

Optimum: $v_1 \to b$, $v_2 \to a$, $v_3 \to a$, value : 3

(b) keep on counter $q_v = 0$ per offline vertex.

On arrival of u, scan $N(u)$: if $q_v < b_v$, assign u to v, $q_v \leftarrow q_v + 1$

otherwise reject

At most $\sum_u |N(u)|$ checks, each on integers of at most input length in bits

Polynomial time, with no vertex copy.

(c) $|O \setminus A| \le \sum_{v \in S} b_v \le |A|$

(d) $OPT = |O| = |O \cap A| + |O \setminus A| \le |A| + |A| = 2ALG$

(e) $b_a = b_b = 1$, $N(u_1) = \{a, b\}$. If reject u_1, stop: $ALG = 0$, $OPT = 1$.

If assign u_1 to $v \in \{a, b\}$, Let $N(u_2) = v$: tit gets 1;

Optimum sends v to the other vertex and u_2 to v, gets 2

∴ No deterministic algorithm guarantees more than $\frac{1}{2}$.